

No urban area will prosper unless it attracts those who can choose to live wherever they wish (Jonathon Barnett)



Bengaluru Commuter Rail Service

But getting here, where we need to be, needs investments

...we get here

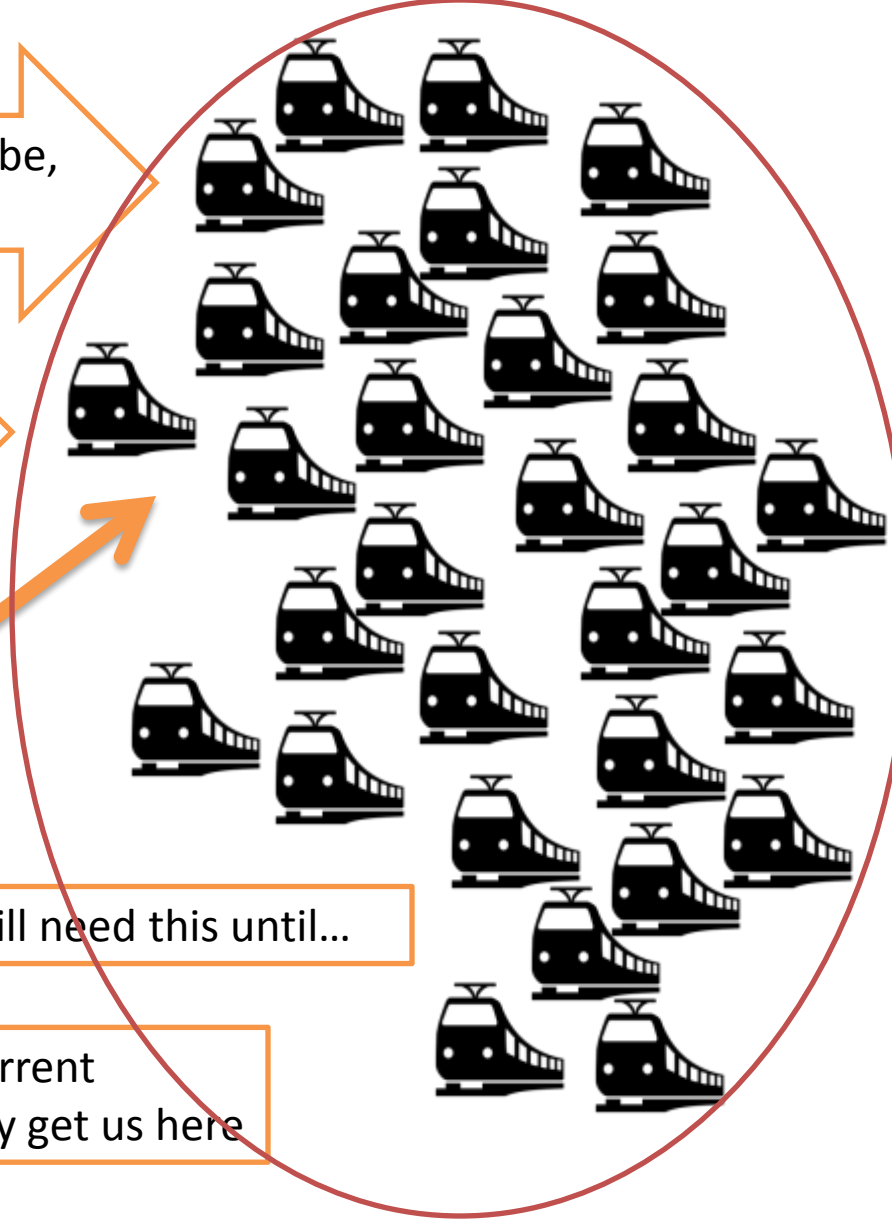
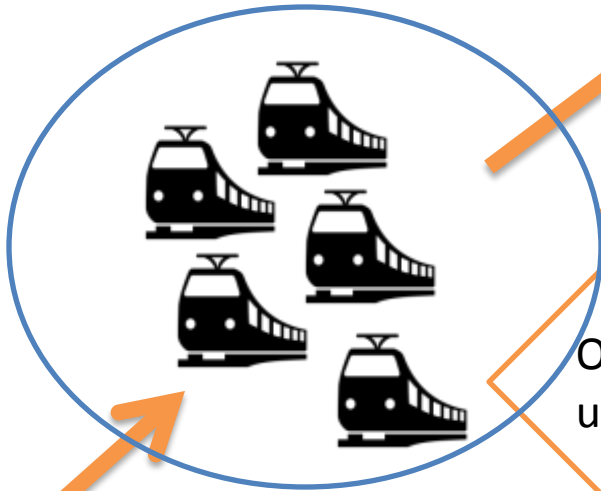
Blue circle = Track optimization

But we still need this until...

Optimizing current utilization may get us here

Red circle = Dedicated Infrastructure

This is our current Commuter Rail System



Typical dialog

We cant give you blue circle because our tracks are utilized

Because we dont intend running commuter rail & the current capacity is sufficient to run our trains for now

We wont get into the red circle because our fare structure will never allow it to be profitable.



**Indian
Railways**



But you never added any reasonable capacity on the 200kms of available tracks?

But this city will collapse without the red circle how can you not run it and also not allow anybody else to run it?

Then we need a model which will make red circle happen

Way forward option

- IR formalizes RVNL JV-SPV guidelines in 2011 budget with BCRS as first pilot
- IR driven
 - Work out commercial model btwn Private parties & IR (GoK may or may not have stake)
- GoK driven
 - GoK forms BCRAI an SPV with IR to execute CRS

The Blue Circle Dialog

BCRS

Projects	Length in Km	Anticipated cost for new line (crores ₹)	Anticipated GOK Share (crores ₹)	Land	GOI share (crores ₹)	Cost per KM (Crores ₹)
Bangalore- Krishnarajpuram- WhiteField	23	85	0	No issue	85	3
Yelahanka- Channasandra	12.9	65	0	Railway has not requested	65	5
Yeshwanthpur- Yelhanka	12	60	0	Railway has not requested	60	5
 Yeshwanthpur- Byappannahalli	20	?	?	?	?	?

*Railways Infrastructure Plan, Dec 2009, IDD GoK

RTI response

$$\text{UIC 406: } K = q_{\max} * n$$

$q_{\max}$ is the maximum traffic intensity [trains/h] & n is the number of train paths

$$K = 5/3 * 2 = \mathbf{3.3 \text{ trains/hour}}$$

SL.N	Stretches	Utilisation of Track between 07.00 to 09.00 A.M.		Utilisation of Track between 05.00 to 08.00 P.M.	
		* Capacity	Frequency	* Capacity	Frequency
(a)		50%	1	50%	1
(b)		100%	3	100%	4
(c)	Yeswanthapur to Hebbal	100%	3	100%	4
(d)	Banaswadi to Anekal	100%	1	100%	1
(e)	Bangalore Cantt. To Whitefield	100%	4	100%	5
(f)	Banaswadi to Whitefield via Baiyappanalli	100%	2	100%	3
(g)	K.R. Puram and Dodaballapur via Yelchanka	100%	1	100%	1
(h)	Chikbanavar to Nelamangala	Not yet opened			

* The utilization of track shown includes trains run in the section other than the trains runs in the originating/ending stretches asked for

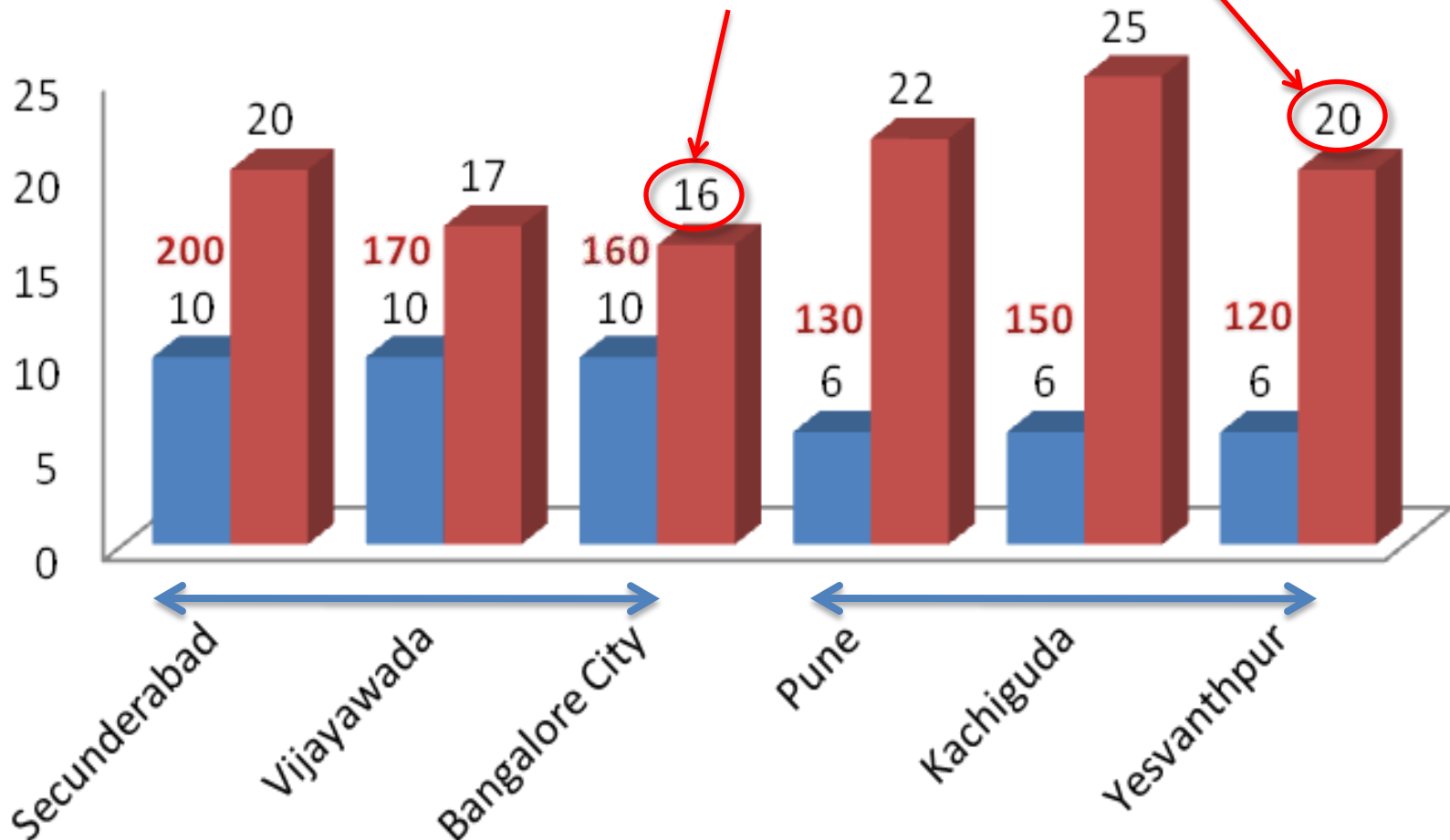
2. The feasibility of Track doubling between Baiyappanahalli-Anekal **has not been done.**

* Evaluation of railway capacity, Aalborg University, Denmark, 2006

Blue circle - Feasibility

- Current most crowded segment BNC-WFD train capacity is **3.3 trains per hour**
- In Japan, on double-track 'main lines', with a mixture of stopping trains, expresses, long-distance services and freight, **minimum** performance benchmark is **15 trains per hour each way** (or one train every four minutes). This applies regardless of the service mix; but if one particular kind of service predominates, then planners expect to be able to run even more trains.
- Hence, **minimum** capacity **available** in SWR tracks is **12 more trains per hour**
- Call to action report asks for **ONE additional train each hour** on all segments as a part of Commuter Rail Service

Lowest Throughput



■ Platforms ■ Platform throughput

*Public sources



Lets Do It -
Fast